ARE STRATOSPHERIC PLATFORMS IN AIRSPACE OR OUTER SPACE?

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Description of Stratospheric Platforms

A stratospheric platform is a structure capable of maintaining its location over a specific portion of the Earth's surface for a multi-year lifetime, at an altitude of approximately 30 kilometers or higher.[1] Such platforms first became practical with the invention and successful demonstration of Corona Ion Engine propulsion technology by Dr. Alfred Wong, Chairman of the Plasma Physics Department at UCLA and Chief Scientist of Sky Station International (SSI).[2] This article is concerned with the application of stratospheric platforms to telecommunications. For this application the stratospheric platforms are equipped with solar power arrays and radio communications equipment. This payload, together with its associated ground infrastructure, enables the stratospheric platform to provide a wide array of wireless telecommunications services, including broadband internet access by portable devices and broadband portable telephony.[3] An application for a global wireless telecommunications network using stratospheric platforms was submitted to the United States Federal Communications Commission (FCC) in early 1996 by SSI.[4]

Benefits of Using Stratospheric Platforms

There are several unique benefits enjoyed by stratospheric platforms as telecommunications relays. These benefits may be summarized as high angles of elevation, short path lengths, low system costs and no environmental impact.

The first benefit is an extraordinarily high angle of elevation with communicators, thus eliminating many losses imposed by terrestrial ground clutter and long paths through rain cells. In a typical configuration, the stratospheric platform will be placed approximately 30

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1 A. Haig, M. Rothblatt, P. Mahon, C. Patusky, J. Blake, L. Tiedrich, *Petition of Sky Station International to Establish a Global Stratospheric Telecommunications System*, Fed. Comm. Comm'n, March 20, 1996 .

2 *Repairing the Ozone Layer*, POPULAR SCIENCE, January 1995, at 21.

3 *Internet Access from Outer Space*, TELEMEDIA WEEK, April 1, 1996, at 54.

4 A. Haig, M. Rothblatt, P. Mahon, C. Patusky, J. Blake, L. Tiedrich, *Application of Sky Station International to for Authority to Construct, Deploy and Operate a Global Stratospheric Telecommunications System*, FCC, March 20, 1996.

kilometers above the target zone of communication. At this altitude, 30 degree or greater angles of elevation will exist across a coverage area of greater than 7500 square kilometers, and angles of elevation will be as high as 90 degrees, on a permanent basis, directly below the stratospheric platform. Between the nadir point and the 30 degree contour, there is a graceful decline of angle of elevation from 90 degrees.

The second benefit is short path lengths, compared to conventional satellite communication systems. The much shorter path length from a stratospheric platform as compared to a conventional satellite platform results in much improved communications margins for an equivalent amount of power. The improved margins facilitate the provision of mobile and portable services and use of vacant frequencies in the millimeter wave band.

A third benefit of stratospheric platforms is a much reduced system cost as compared to satellite networks. Satellite capital costs consist in large part of satellite construction costs, launch costs, and insurance costs. The insurance costs are at least 15% of the sum of satellite construction and launch costs, reflecting the historical failure rates of launching satellite communications systems. Launch costs are, to a first approximation, roughly equal to satellite construction costs. Stratospheric platforms do not require rocket launches since they "float" to location under helium pressure and Corona Ion Engine propulsion. Insurance costs are likely to be negligible since the past 50 years of lighter-than-air craft has been one of a stellar safety record. Accordingly, assuming that stratospheric platforms cost approximately the same as a satellite platform, the stratospheric system cost will be approximately 43% of the satellite system cost. The actual total system cost of a stratospheric system or a satellite system must also consider the number of platforms deployed and the replacement lifetime of each platform. Such calculations and comparisons are beyond the scope of this paper.

A final benefit of stratospheric platforms is a reduced environmental impact as compared to rocket-launched platforms. Large networks of low earth orbit satellites, especially if deployed using solid fuel rockets, have a deleterious effect on the earth's ozone layer.[5] By comparison, the stratospheric platforms, using ambient ions in the stratosphere and neutral helium for buoyancy, present no adverse

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[5] "Unfortunately, space programs have themselves been linked to ozone loss. Global industries reach into outer space through the launching industry. Ozone is depleted through solid - but not liquid - fueled rockets. The exhaust often contains hydrochloric acid and the chlorine therein breaks down ozone molecules. The obvious solution is to move from solid back to liquid fuels. This is the recommendation of the March 1990 panel of the National Research Council of the National Academy of Sciences." J. Galloway, *Protecting the Ozone Layer: The 1990 London Revisions to the Montreal Protocol*, in 34 PROC. COLLOQ. L. OUTER SPACE 177, 181 (1992). *See also*, William J. Broad, Some Say the Rocket's Red Glare is Eating Away the Ozone Layer, NEW YORK TIMES, May 14, 1991, B11; Carl Q. Christol, *Stratospheric Ozone, Space Objects and International Environmental Law*, 4 J. SPACE L. 23 (1976).

environmental impact. Due to the fragile nature of the earth's ozone layer, and the proven connection with increased skin cancer and other health disorders from its depletion[6], it would appear reasonable to use non-pollutive telecommunications systems wherever possible.

Regulatory Issues Presented by Stratospheric Platforms

Stratospheric platforms present three categories of regulatory issues. The first set of issues concerns the jurisdictional competence for stratospheric platforms. Are they in a nation's sovereign airspace or are they in an international *res communes* realm? The second set of issues concern the appropriate telecommunications regulatory framework. Are stratospheric platforms part of the "space services" or all the other services, including terrestrial services, maritime services, and aeronautical services? The third set of issues relates to deployment questions. Which country or countries are empowered to authorize deployment of the platforms, the ground stations, and the telecommunications services?

The Legal Status of the Stratosphere

It often comes as a surprise that there is no international legal definition concerning where airspace ends and outer space begins. This fact is surprising because the legal regime is very different in airspace and outer space. Airspace is subject to the national sovereignty, and hence the laws, of the nation lying vertically below. Outer space is, by international treaty, not subject to national sovereignty by claim of right or use.[7]

Debates concerning the delimitation of airspace and outer space occur within a venue called the United Nations Committee on the Peaceful Uses of Outer Space (COPUOS). This committee has both a Technical Subcommittee and a Legal Subcommittee. Despite over a decade of study of the matter, neither Subcommittee, nor the parent Committee, has ever agreed on a definition of a boundary between airspace and outer space. For example, in the 1986 Report of the Chairman of the COPOUS Working Group concerned with definitional matters it was said that a number of delegations felt "the establishment of a boundary between airspace and outer space would also be arbitrary as there was no clear scientific basis for fixing at a particular altitude the demarcation between airspace and

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[6] *See generally, Stratospheric Ozone Depletion, Subcommittees on Hazardous Wastes and Toxic Substances and Environmental Protection of the Committee on Environment and Public Works*, U.S. Senate, Joint Hearing, 100th Cong., 2d Sess., March 30, 1988

[7] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, Jan. 27, 1967, 18 U.S.T. 2410, T.I.A.S. No. 6347, 610 U.N.T.S. 205 (entered into force Oct. 10, 1967).

outer space." It was further observed that "an unnecessary definition or delimitation of outer space would be undesirable and might cause difficulties, including interference with the development of space activities, not hitherto encountered. Such a premature definition and delimitation might also prove inappropriate in the light of continuing technological advances."[8] And as recently as March 1995, the United States advised the COPOUS that "the lack of certainty as to where airspace ends and outer space begins has not impeded the development of activities in either sphere, and we fear that the premature injection of unnecessary law would have precisely that unfortunate effect."[9]

The principal proponent at the UN of a specific boundary between airspace and outer space was the Soviet Union. For several years it had formally submitted proposals seeking a specific boundary.[10] However, since the breakup of the Soviet Union, no further such formal proposals have been forthcoming. The Soviet Union's proposal had been that the boundary be set at around 100-110 kilometers, that being the lowest perigee achieved by orbiting satellites. The Soviet Union's view was countered by other countries because there were satellites whose perigee dipped as low as 90 kilometers, there were rocket-assisted air vehicles that climbed as high as 100 kilometers for brief periods, and the historical practice of air flight and space flight may not be relevant to what the air/space boundary should be.[11]

At the most recent meeting of the COPUOS Legal Subcommittee, during April 1996, there was still no consensus whatsoever as to what the boundary between airspace and outer space was, and whether setting such a boundary was advisable.[12] Among the concerns raised were that the UN

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[8] U.N.G.A., Committee on the Peaceful Uses of Outer Space, Report of the Legal Sub-Committee on the Work of its Twenty-Fifth Session (24 March-11 April 1986)), at 20.

[9] U.N.G.A., Committee on the Peaceful Uses of Outer Space, Legal Sub-Committee, Thirty-Fourth Session, March 1995, Statement by U.S. Alternative Representative, Mr. Jeffrey Maughan Maclure.

Referring to the application of Sky Station International to the FCC to create a new Global Stratospheric Telecommunications Service (GSTS) by using a revolutionary technology that holds each of the proposed 250 Sky Station platforms stationary at a 3 km altitude, Professor Gorove states that "this development suggests the necessity of exercising continued caution to avoid premature determination of demarcation lines." *See* Stephen Gorove, *The Growth of Space Law through the Cases*, 24 J. SPACE L. 4 (1996).

[10] *See supra* note 8.

[11] *Id.*

[12] U.N.G.A., Committee on the Peaceful Uses of Outer Space, Report of the Legal Subcommittee on the Work of its Thirty-Fifth Session (18-28 March 1996), Sec. II, "Matters Relating to the Definition and Delimitation of Outer Space and to the Character and Utilization of the Geostationary Orbit, Including Consideration of Ways and Means to Ensure the Rational and Equitable Use of the Geostationary Orbit Without Prejudice to the Role of the International Telecommunication Union."

should discontinue its "unproductive debate on [whether] the direct and topographical or indirect and functional approach to the definition and delimitation of outer space" is best, that aerospace objects did not lend themselves to be objects which traversed arbitrarily defined boundaries, and that a "functional boundary had been generally acquiesced upon since no State had objected to flights of satellites over its territory."[13]

In summary. after more than ten years of debate, it can be said clearly that: (1) there is no legal boundary between airspace and outer space, and (2) there is no near term likelihood that such a boundary will be established. With regard to stratospheric platforms the situation is that if several years go by and countries worldwide generally have no objection to stratospheric platforms, then by force of common international practice they will be deemed to be in a region of outer space. On the other hand, if each country over which a stratospheric platform is located demands that it authorize the stationing of a stratospheric platform, then by force of common international practice they will be deemed to be in a region of airspace.

Experts have also long opined on the subject of where the airspace ends and outer space begins. Perhaps the greatest treatise ever written on the law of space is the mammoth 1200 page LAW AND PUBLIC ORDER IN SPACE by Profs. Myres S. McDougal, Harold D. Lasswell and Ivan A. Vlasic of Yale University.[14] They note in the treatise: "One of the most often asked questions is, "Where does outer space begin? The briefest, if not entirely satisfactory, reply is that there is no answer which will serve everybody -- that much depends upon who is making the classification for what purposes, and that, therefore, there is more than one answer."[15] After a survey of all the different possible answers -- ranging from the 12 mile limit (60,000 feet) at which the human body begins to boil and medical scientists have designated the "space-equivalent altitude" to the 600 mile limit at which collisions between air particles are extremely rare and space scientists have designated the "exosphere" or the "astronomical material frontier" -- the professors concluded: "This survey should serve to indicate the hopelessness of attempts to determine "scientifically" a boundary between "outer space" and "airspace" which would be both precise and valid for all purposes and every contingency. Different sciences have different criteria for defining "airspace" and "outer space" and there obviously cannot exist an all-purpose definition that would satisfy the unique requirements of them all. This led an eminent space scientist to conclude that the 'scientific agreement on where space begins is a self-evident impossibility.' Even the boundaries suggested by various scientists for their own special disciplines appear to be subject to change, as new knowledge becomes available."[16]

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13 *Id.* at 15.

14 MYRES S. MCDOUGAL, HAROLD D. LASSWELL, IVAN A. VLASIC, LAW AND PUBLIC ORDER IN SPACE (Yale U. Press, 1963).

15 *Id.* at 33.

16 *Id.* at 35.

Finally, it might be asked, where *should* the boundary between airspace and outer space be set, *i.e.*, which boundary *would be in the best interests of the peoples of the world?* From a legal standpoint, it is fairly clear that including the stratosphere within the realm of outer space is a more progressive outcome than including it within airspace. There are two main reasons for this conclusion. If the stratosphere is deemed to be part of outer space, then activities in the stratosphere will fall within the purview of the Outer Space Treaty of 1967.[17] This Treaty has a number of progressive provisions which are in the best interests of the world's population. First, the Treaty determines that any damage that occurs to the surface of the earth, or to aircraft in flight, from an outer space object is based on absolute liability.[18] In comparison, damage caused by objects in air space is compensable only of the object in air space was "at fault." Hence, should a stratospheric object ever cause any harm to the earth, the damaged persons will be able to collect compensation more easily, and without having to prove fault, if the stratosphere is deemed to be part of outer space.

Second, the Treaty provides that any object placed in outer space must be used only for peaceful purposes, must be notified to the United Nations and must involve the participation of all countries irrespective of their stage of economic development.[19] Hence, if the stratosphere is part of outer space it will more likely be a region of peaceful development rather than "war games," all countries will have more information about what is going on in the stratosphere, and all countries will be able to better reap economic rewards from stratospheric activities. On the other hand, if the stratosphere were deemed to be part of airspace, then there is no obligation of a country to keep its stratospheric activities peaceful, no obligation of a country to inform even neighboring countries of its stratospheric activities, and no obligation of any country to share the benefits of stratospheric activities with developing countries. Rather clearly, the

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[17] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, Jan. 27, 1967, 18 U.S.T. 2410, T.I.A.S. No. 6347, 610 U.N.T.S. 205 (entered into force Oct. 10, 1967).

[18] Convention on International Liability for Damage Caused by Space Objects, opened for signature March 29, 1972, 24 U.S.T. 2389, T.I.A.S. No. 7762 (entered into force Oct. 9, 1973), explained in the context of aerospace vehicles in M. Rothblatt, *International Liability of the United States for Space Shuttle Operations*, 13 INT'L LAW. 471 (1979). It should be noted that under the Convention on International Liability, citizens can not sue their own government for damage caused by space object. However, if stratospheric platforms are in outer space, and hence space objects, any domestic harm they cause can still be redressed using national laws instead of the Convention on International Liability.

[19] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, Jan. 27, 1967, 18 U.S.T. 2410, T.I.A.S. No. 6347, 610 U.N.T.S. 205 (entered into force Oct. 10, 1967), Articles I, II and III.

much more progressive result is for the stratosphere to be deemed to be part of outer space, and hence subject to the Outer Space Treaty of 1967.

It should also be noted that a country gains very little, if any, national sovereignty protection from deeming the stratosphere to be part of airspace. Probably the greatest fear is that of telescopic observations. However, telescopic observations that have extraordinary resolution (good enough to see a tennis ball) can be performed from satellites in low earth orbit. Hence, prohibiting stratospheric platforms for observation-related reasons accomplishes nothing. Another reason to exert national sovereignty might be to ensure the safety of the stratospheric platform. However, this objective can be accomplished as a quid-pro-quo for authorizing telecommunications service rights into a country, to the extent it is a concern, and does not require the extreme step of extending national sovereignty into the stratosphere.

In summary, the stratosphere is not part of airspace as legally defined today, because there is no legal boundary for airspace. It will not be possible to get scientists to agree on a boundary between airspace and outer space. The stratosphere is above what scientists call "space-equivalent altitude," and is far above the altitude at which countries seek and obtain overflight permission. Finally, the people of the world will be better off if the stratosphere is deemed subject to the Outer Space Treaty of 1967. For all of these reasons, and the paucity of any counterarguments, it is believed that stratosphere platforms should be considered as in outer space.

Do Stratospheric Platforms Provide Space Services?

The International Telecommunication Union (ITU) and most national authorities divide radio communication systems into two main categories -- space services and terrestrial services. Different regulatory obligations attach to each general kind of service. The ITU defines Space Radiocommunication as "Any radiocommunication involving the use of one or more space stations or the use of one or more reflecting satellites or other objects in space."[20] Terrestrial Radiocommunication is defined as "Any radiocommunication other than space radiocommunication or radio astronomy."[21] In other words, if it is not a space service, then it is a terrestrial service, so it is key to see whether a stratospheric platform is a "space station," or a "reflecting satellite" or an "object in space." In fact, it will not be necessary to go further than the definition of space station, which the ITU says is "a station located on an object which is beyond, is intended to go beyond, or has been beyond, the major portion of the Earth's atmosphere."[22] It is well accepted by the entire scientific community that

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[20] International Telecommunication Union, Radio Regulations, (Geneva 1979 - 1995), Art. 8.

[21] *Id.*

[22] *Id.*

97% of the mass of the atmosphere is found below 30 kilometers altitude.[23] Hence, it is incontrovertible that a stratospheric platform is beyond "the major portion of the Earth's atmosphere," and is therefore a "space station" within the framework of the ITU. Since the stratospheric platform is a space station, any telecommunications services that it provides are space radiocommunication services.

It should also be noted that under the ITU's rules, stratospheric platforms are not "satellites" because satellite is defined as "a body which revolves around another body of preponderant mass and which has a motion primarily and permanently determined by the force of attraction of that other body."[24] Since the stratospheric platforms do not revolve around the earth, they are not satellites. This point is important because frequencies within the ITU are allocated to specifically defined sub-services, such as Fixed Satellite Service, Mobile Satellite Service, Broadcast Satellite Service, and so on. While Fixed Satellite Service is defined as a space radiocommunication service "between earth stations at given positions, when one or more satellites are used," Mobile Satellite Service is defined as a space radiocommunication service "between mobile earth stations and one or more space stations." Hence, the Fixed Satellite Service must involve a "satellite" while the Mobile Satellite Service (and all other satellite services defined by the ITU) may involve a "space station." Due to the use of millimeter wave frequencies, stratospheric platforms will unavoidably be in the Fixed Satellite Service because propagation conditions are too severe to engineer for reliable mobile reception. In practical use, all parties within the ITU treat the words "satellite" and "space stations" as synonyms.

In point of fact, the word "satellite" is an anachronism due to the fact that the Fixed Satellite Service was defined as Communications Satellite Service at the ITU's Extraordinary Administrative Radio Service in 1963 while all of the other space radiocommunications services were defined at the ITU's World Administrative Radio Conference for Space Telecommunications in 1971. Between 1963 and 1971 the ITU's experts had grown to appreciate the benefits of having a parallel linguistic structure between "earth stations" and "space stations," and hence from 1971 onwards all space services were defined in terms of space stations and earth stations, leaving as a historical artifact the use of the word "satellite" in the definition of the Fixed Satellite Service. However, whether or not a modification occurs, the artifactual inclusion of the word satellite in the definition of Fixed Satellite Service should in no way impede the use of Fixed Satellite Service frequencies by stratospheric space stations because, as mentioned, in everyday practice, the terms satellite and space station are synonymous.

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[23] S. H. LAY & H. TAUBENFIELD, THE LAW RELATING TO THE ACTIVITIES OF MAN IN SPACE 6 (1970).

[24] *See also*, Fed. Comm. Comm'n, 47 CFR Sec. 2.1 (1996).

National Authorization of Stratospheric Networks

Insofar as stratospheric networks are defined within the ITU as space radiocommunications services, national authorization procedures would be expected to be no different than for other space radiocommunications services. Specifically, any administration may authorize one or more stratospheric space stations, with earth coverage patterns than extend beyond their own territory, provided that the stratospheric space stations are technically coordinated with any other country using the same frequency bands under ITU procedures. The stratospheric space stations may only communicate with users on the ground in countries which have authorized such users to communicate with the stratospheric space stations. Generally, a country issues a blanket regulation authorizing all type-accepted communications equipment designed for use with a particular space service to be sold. In addition, each country which has a ground station that interconnects with the stratospheric space station, for purposes of tie-in with the internet and other PSTN services, would also specifically authorize each ground station. Such permits are usually granted routinely upon the application to the Ministry of Telecommunication from the intended ground station operator.

Conclusion

Stratospheric platforms herald a new age of communications. Such platforms have some of the best attributes of satellite systems, such as wide area coverage, and some of the best attributes of terrestrial systems, such as short path lengths. From a regulatory standpoint, the stratospheric networks are Fixed Satellite Service systems operating under an outer space legal regime. Earth stations accessing the stratospheric platforms, both gateway ground stations and handheld user terminals, will need to be authorized by each country that wants to participate in the global stratospheric network.